

Supplementary Information

Unconcentrated markets, persistent ties: incumbency in Türkiye’s public IT procurement

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Contents

S1 Data construction	1
S1.1 Source and IT scope	1
S1.2 Firms, buyers, and markets	1
S1.3 Descriptive statistics by buyer sector	2
S1.4 Article 21 of Law No. 4734	3
S2 Concentration	3
S3 Incumbency and null models	4
S3.1 Subgroups and sensitivities	4
S3.2 Nulls that preserve firm geography	4
S3.3 Named-product licences and maintenance	5
S3.4 Renewals	5
S3.5 Buyer dependence	6
S4 Supplier choice and state dependence	6
S4.1 Simulated worlds	8
S5 Procedure and timing	10
S5.1 Contract-level model	10
S5.2 Event study	10
S5.3 Dyad continuation	10
S6 Single bidding	11
S7 Estimated costs and discounts	12
S8 Network structure	13

S1 Data construction

S1.1 Source and IT scope

Award records were collected from EKAP (ekapv2.kik.gov.tr) by searching domain keywords. Search matches substrings of tender titles, which admits unrelated tenders. The keyword *erp* retrieved 618 contracts, of which 546 are not IT (it matches, among others, the place name Yaverpaşa and *tuz serpici*, a salt spreader). The phrase *teknik destek* retrieved 1,186 contracts, of which 798 are hospital or university staffing and cleaning contracts. An earlier count-based cleaning had removed 394 obvious false positives from 13,418 records, leaving 13,024.

We re-audited this set by value. All 150 contracts with the highest value were classified by hand (108 IT, 1 call-centre service, 19 mixed, 22 not IT). For the remainder, ordered title rules assign each

contract to one of four scope classes: IT, IT-enabled call-centre services, ambiguous, and not IT. Works contracts are ambiguous only if they install an IT or operational-technology system, such as city security cameras, SCADA, fibre, or a data centre, and are otherwise not IT. Staffing contracts for data entry are ambiguous; bundled cleaning or catering contracts without an IT system are not IT. Random samples from every rule were inspected, and seven rule outcomes for top contracts were overridden by hand. Table S1 reports the result. The main sample consists of the IT and call-centre classes.

EKAP’s award list has one row per tender identifier, so tenders with several lots appear with a single winner. We found 116 main-sample titles that suggest multiple lots and no framework-agreement call-offs (the word *çerçeve* appears only in the sense of “within the framework of”).

Table S1: Scope classes after the value-based audit (13,024 contracts that passed the earlier count-based cleaning).

Scope class	Contracts	Nominal value (bn TRY)	Share of value (%)
IT	9,981	25.64	56.5
IT-enabled call-centre services	10	7.62	16.8
Ambiguous (excluded from main sample)	1,406	4.12	9.1
Not IT (removed)	1,627	8.03	17.7
Total	13,024	45.42	100.0

S1.2 Firms, buyers, and markets

Firms. Names were canonicalized by upper-casing with Turkish locale, removing punctuation and whitespace, and stripping legal-form suffixes from the end of the name, subject to a minimum remaining length. Joint ventures were kept separate from their members. The merge groups of the roughly 300 largest firms were reviewed by hand. Seventy-two brand groups were confirmed, including firms that changed legal form (for example from limited to joint-stock company) or name while keeping the same brand in non-overlapping years. Three uncertain brand pairs were left unmerged. The process reduces 4,788 recorded names to 4,389 firms. By a name-pattern heuristic, 536 of these firms (302 in the main sample) are natural persons. In the released data, every winner that is a natural person or lacks an explicit legal-entity form (690 firms) is replaced by a stable pseudonym.

Buyers. Thirty-six generic buyer labels pool units from two or more provinces, among them the provincial health directorates and hospital unions of the Ministry of Health. We split them by the province recorded for each tender (1,001 of the 13,024 contracts; 34 labels and 732 contracts in the main sample), which turns 3,453 labels into 3,663 buyer nodes. Six national bodies with offices in two provinces were not split. The recorded labels are used as a sensitivity check throughout.

Buyer sectors. Buyers were assigned to 11 sectors by ordered keyword rules on the buyer name. Relative to an earlier coding, 1,423 contracts changed sector, mainly because upper-casing without Turkish locale had prevented mixed-case names from matching, and because several rules matched unintended substrings.

Product markets. Contracts were assigned to 14 product markets by ordered rules on the tender title, most specific first; in the main sample, 97% receive a specific market. To validate the rules, a random sample of 300 main-sample titles (seed 7) was labelled by three AI coders working blind to the rules: an earlier check and two independent coders based on different language models, each seeing only the titles and a codebook. The two independent coders agree on 88.7% of titles ($\kappa = 0.87$), and each agrees with the earlier check on 90–92% ($\kappa = 0.89$ – 0.91). The rules agree with the independent coders on 73–75% of titles ($\kappa = 0.70$ – 0.72), and with their joint label on 79.3% of the 266 titles where the two coders agree. Against that joint label, health information systems has precision 1.00 and recall 0.99, GIS 1.00 and 0.94, and software licences 0.80 and 0.97. The weakest classes are custom software (0.48 and 0.56), maintenance (0.60 and 0.41), and the residual “other

IT” class. The labels are AI-produced and have not been checked by human coders; blind coding sheets for two human coders are part of the replication material.

Dates, procedures, values. Dates are tender dates joined by tender identifier (coverage 100%, October 2010 to March 2026). Procedures are open, restricted, and the negotiated procedures of Article 21(a)–(f); the extract contains no direct procurement under Article 22. Values are deflated to 2025 lira with the annual average of the consumer price index (2003=100; 2026 from the linked monthly series). With the domestic producer price index (Yİ-ÜFE) instead, count-based indices are unchanged. Seven single-variant value classifications move across the 1,000 or 1,800 boundary, and no robust classification changes.

S1.3 Descriptive statistics by buyer sector

Table S2: Buyer sectors (main sample). Value in bn 2025 TRY; procedure shares in % of contracts.

Buyer sector	Contracts	Firms	Buyers	Value	21(b)	21(f)	Open
Health	2,950	554	1,098	30.93	28.5	9.4	61.8
Municipal / local	2,688	1,084	667	19.93	2.2	44.1	53.4
Other public	1,295	646	351	10.88	2.5	41.1	54.7
Education	1,054	519	330	4.68	5.1	31.8	62.6
Infrastructure / transport	644	337	123	7.72	2.8	39.8	57.0
Agriculture / environment	395	231	98	4.29	2.3	31.4	63.5
Social / finance	303	156	42	5.20	3.6	39.3	55.4
Provincial administration	223	125	74	0.99	1.3	25.1	62.8
Security / interior	202	142	64	1.50	3.5	26.2	66.8
Defence	191	96	29	1.89	1.0	3.7	95.3
Justice	46	32	14	1.25	26.1	28.3	43.5
All	9,991	2,814	2,890	89.26	10.5	29.6	59.0

S1.4 Article 21 of Law No. 4734

Translations are ours. The chapeau reads: “The negotiated procedure may be used in the following cases.”

- (a) No tender was submitted in an open or restricted procedure.
- (b) Before 25 May 2018: the tender must be held urgently because of sudden and unexpected events such as natural disasters, epidemics, or a danger of loss of life or property, or of events the contracting authority could not foresee. Law No. 7144 (Art. 11, in force 25 May 2018) inserted: “or in cases that have special features in terms of construction technique, or in which the contracting authority determines that the work must be carried out urgently to ensure structural safety or the safety of life and property”.
- (c) The tender must be held urgently because of special defence- or security-related circumstances.
- (d) The procurement requires research and development and is not subject to serial production.
- (e) The technical and financial characteristics cannot be specified with the necessary precision because the goods, services, or works are original and complex.
- (f) Purchases of goods, materials, or services whose estimated cost is below a monetary limit, updated every 1 February (for example 157,923 TRY in 2014 and 3,406,508 TRY in 2026).

In cases (b), (c), and (f), no notice needs to be published; at least three firms are invited to submit qualification documents and price offers. In our data all 2,957 main-sample 21(f) contracts are goods or services, and from February 2014 on, 99.96% are below the limit in force.

S2 Concentration

Table S3 reports supplier concentration by product market. Bootstrap intervals are bias-shifted percentile intervals (1,000 resamples of contracts within market). “No max” removes the market’s largest contract; “winsor.” caps contract values at the 99th percentile of the main sample. Table S4 reports medians of HHIs computed within calendar years for market-years with at least 30 contracts. The bias-corrected count index is the unbiased estimator $10,000 \sum_i n_i(n_i - 1)/[n(n - 1)]$, where n_i is firm i ’s number of contracts and n is the market-year total.

Table S3: Supplier concentration by product market (main sample; HHI on 0–10,000).

Product market	Contracts	Firms	HHI count [95% CI]	HHI value [95% CI]	Value, no max	Value, winsor.	CR4 value (%)
Health information systems	2,189	205	431 [405, 458]	831 [692, 985]	868	811	53
ERP / management software	1,515	599	84 [71, 97]	132 [73, 219]	120	96	16
Custom software / web / mobile	1,117	734	25 [22, 29]	264 [41, 744]	176	110	27
Software licences	1,047	375	103 [89, 118]	275 [188, 441]	276	270	25
Network / data-centre infrastructure	748	390	72 [60, 87]	281 [151, 523]	234	226	28
Computers / peripherals	696	467	40 [34, 47]	170 [69, 363]	146	155	18
Maintenance / support services	626	297	82 [69, 98]	301 [196, 481]	303	296	25
GIS / city information	553	203	353 [273, 444]	408 [228, 842]	385	347	33
Cybersecurity	390	194	193 [143, 263]	476 [193, 1,002]	442	340	37
Physical security / surveillance	385	222	84 [69, 104]	212 [123, 373]	208	210	20
Other IT	277	215	79 [58, 111]	767 [0, 2,275]	603	296	46
Education technology	205	128	225 [147, 346]	1,062 [196, 2,936]	752	537	54
Smart city / traffic / OT	169	106	171 [125, 234]	2,580 [503, 5,917]	823	568	66
Call centre / help desk	74	46	690 [363, 1,160]	8,908 [3,534, 10,000]	1,685	1,391	98
All markets pooled	9,991	2,814	39 [36, 41]	149 [40, 504]	92	87	18

Table S4: Median within-year supplier HHI by product market.

Product market	Years	Count HHI	Count HHI, bias-corr.	Value HHI	Value HHI, no max
Health information systems	16	671	597	1,091	1,081
GIS / city information	12	650	369	1,503	1,206
Physical security / surveillance	4	427	108	1,098	926
Cybersecurity	7	416	156	1,666	1,263
Maintenance / support services	14	343	91	1,627	1,244
Software licences	15	325	187	1,029	868
Network / data-centre infrastructure	15	274	68	1,038	937
Computers / peripherals	15	265	44	903	762
ERP / management software	15	226	119	688	521
Custom software / web / mobile	15	176	37	743	601
All markets pooled	17	69	55	256	209

Decomposition. The pooled index satisfies $H_{\text{agg}} = \sum_m w_m^2 H_m + C$, where w_m is market m ’s share and $C = \sum_i \sum_{m \neq m'} \text{sim}_{im'} w_m w_{m'}$ collects the terms of firms active in several markets. By count, $H_{\text{agg}} = 38.6$, $\sum_m w_m^2 H_m = 26.7$ and $C = 11.9$; $\sum_m w_m^2 = 0.117$ and the demand-weighted mean of market HHIs is 179. By real value the corresponding figures are 148.6, 127.9 and 20.7, with a weighted mean of 1,278 (534 without the largest contract). A quarter of firms are active in more than one market; they hold 69% of contracts.

S3 Incumbency and null models

S3.1 Subgroups and sensitivities

Definitions are given in the main text. Tables S5–S9 report incumbency by subgroup against the same permutation draws restricted to the subgroup. The nulls are N1 (market $\times$ year), N2 (adding buyer sector), and N3 (adding buyer province instead). All subgroup tests under N1 have $p < 0.001$, except for four very small groups: restricted procedure ($p = 0.24$, $n = 8$), Article 21(a) ($p = 0.04$, $n = 7$), and Article 21(d) and the year 2010 (one eligible contract each). Table S10 reports sensitivities; Figure S1 shows incumbency by year.

Table S5: Incumbency by product market.

Product market	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health information systems	1,252	0.716	0.151 [0.136, 0.167]	0.157	0.407
Maintenance / support services	375	0.536	0.029 [0.013, 0.045]	0.139	0.122
Physical security / surveillance	182	0.418	0.030 [0.011, 0.055]	0.097	0.317
Other IT	112	0.393	0.019 [0.000, 0.045]	0.132	0.233
ERP / management software	745	0.392	0.027 [0.016, 0.039]	0.077	0.111
Network / data-centre infrastructure	328	0.384	0.023 [0.009, 0.040]	0.106	0.121
Smart city / traffic / OT	58	0.397	0.041 [0.000, 0.103]	0.104	0.314
Cybersecurity	161	0.398	0.052 [0.019, 0.087]	0.132	0.186
Computers / peripherals	281	0.327	0.012 [0.000, 0.025]	0.047	0.164
Software licences	602	0.334	0.039 [0.025, 0.055]	0.105	0.139
Education technology	80	0.350	0.068 [0.025, 0.125]	0.201	0.168
Call centre / help desk	19	0.316	0.062 [0.000, 0.158]	0.222	0.178
GIS / city information	230	0.326	0.088 [0.057, 0.122]	0.117	0.146
Custom software / web / mobile	488	0.201	0.006 [0.000, 0.014]	0.029	0.056

Table S6: Incumbency by buyer sector.

Buyer sector	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health	1,502	0.667	0.129 [0.116, 0.142]	0.143	0.372
Other public	583	0.395	0.030 [0.017, 0.045]	0.075	0.120
Education	439	0.364	0.028 [0.014, 0.043]	0.118	0.164
Municipal/Local	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Provincial admin	115	0.374	0.046 [0.009, 0.087]	0.108	0.309
Social/Finance	196	0.357	0.041 [0.015, 0.066]	0.185	0.086
Agriculture/Environment	210	0.348	0.042 [0.019, 0.067]	0.110	0.100
Infrastructure/Transport	342	0.322	0.029 [0.015, 0.047]	0.097	0.116
Defence	131	0.298	0.039 [0.008, 0.076]	0.219	0.070
Security/Interior	100	0.280	0.025 [0.000, 0.060]	0.145	0.103
Justice	24	0.292	0.038 [0.000, 0.125]	0.292	0.091

Further robustness checks give the following results. Excluding the 116 possible multi-lot tenders gives an observed incumbency of 0.455 (N1/N2/N3 nulls 0.062/0.109/0.209). Excluding natural-person winners gives 0.464 (0.066/0.116/0.214). The excess amounts to 1,919, 1,683, and 1,203 incumbent contracts under N1, N2, and N3.

S3.2 Nulls that preserve firm geography

A firm's home province is the province where it won most contracts before the tender; firms without earlier wins form their own category. Null N4 permutes winners within market $\times$ year $\times$ winner's home province: the null mean is 0.173 (ratio 2.61, excess 1,370 contracts). Null N4b permutes within market $\times$ year $\times$ buyer province $\times$ winner's home province: the null mean is 0.264 (ratio 1.71, excess 923 contracts). Under N4b, 38.5% of eligible contracts are in cells with a single possible winner, which makes it very conservative. Renewals have an incumbency of 0.720 against an N4b null of 0.391, new needs 0.305 against 0.195 (all $p < 0.001$). A version with home provinces computed from all wins, including the contract itself, gives null means of 0.185 (N4) and 0.242 (N4b).

S3.3 Named-product licences and maintenance

Titles that indicate licence renewal, maintenance, support, or updates of a named product (for example hospital information systems, Oracle, Microsoft) account for 10.5% of repeat-eligible contracts. Their incumbency (0.486) is close to that of other contracts (0.448). Excluding them, renewals have an incumbency of 0.723 against an N3 null of 0.296 (2.44 times), and new needs 0.306 against 0.163

Table S7: Incumbency by buyer type (keyword classes of buyer names).

Buyer type	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
MoH hospital / dental centre	617	0.831	0.177 [0.154, 0.203]	0.181	0.386
University hospital / medical faculty	110	0.627	0.099 [0.045, 0.155]	0.120	0.363
MoH provincial directorate / hospital union	695	0.563	0.098 [0.079, 0.118]	0.113	0.391
University (non-health unit)	401	0.362	0.025 [0.012, 0.040]	0.113	0.164
Municipality / municipal company	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Other	296	0.351	0.031 [0.014, 0.051]	0.086	0.183
Other health body (central/agency)	82	0.378	0.058 [0.012, 0.110]	0.139	0.130
Central government / agency	1,441	0.353	0.035 [0.027, 0.045]	0.124	0.107

Table S8: Incumbency by procedure.

Procedure	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Article 21(b)	618	0.723	0.143 [0.120, 0.165]	0.159	0.376
Article 21(c)	34	0.412	0.020 [0.000, 0.088]	0.069	0.255
Open	3,041	0.441	0.057 [0.051, 0.064]	0.112	0.203
Article 21(f)	1,204	0.346	0.033 [0.023, 0.043]	0.080	0.131
Article 21(a)	7	0.286	0.049 [0.000, 0.286]	0.173	0.227
Restricted	8	0.125	0.033 [0.000, 0.125]	0.059	0.035
Article 21(d)	1	0.000	0.001 [0.000, 0.000]	0.000	0.000
Article 21(e)	0	–	– [–, –]	–	–

(1.88 times). Excluding all titles with any licence, maintenance, support, or update wording (44% of eligible contracts) leaves ratios of 2.07 and 1.77.

S3.4 Renewals

Titles are normalized by Turkish-aware lower-casing, removal of diacritics, digits, month names, and about 110 generic procurement words, and stemming to five characters; similarity is the Jaccard index of the stemmed word sets. A contract is a renewal if the buyer has a contract in the same market 183–548 days earlier with similarity of at least 0.5. Sixty best-match pairs, drawn at random within four similarity bands (15 each), were coded blind by the same AI coders. Of the 42 pairs above the threshold, 35–40 were judged the same recurring service (depending on the coder), but so were 6–10 of the 18 pairs below it. Agreement between the rule and the coders is moderate ($\kappa = 0.40$ – 0.45), and between the two independent coders $\kappa = 0.63$. The rule is thus precise but misses renewals whose titles were reworded, so the new-need group contains some renewals. Table S11 reports results by procedure and market and alternative definitions.

S3.5 Buyer dependence

Figure S2 shows the distribution of main-supplier shares. With per-buyer permutation p -values and Benjamini–Hochberg control of the false discovery rate at 5% across the 540 buyers, 273 buyers (50.6%) exceed their N1 null on the main-supplier share and 357 (66.1%) on the supplier HHI; under N2 the figures are 212 (39.3%) and 289 (53.5%). With 1,000 permutations, no buyer can pass a Bonferroni correction. The share of buyers above their 95th null percentile (uncorrected) rises with buyer size (50% for 5–9 contracts, 63% for 10–19, 80% for 20 or more). The mean excess, however, falls with size (0.24, 0.23, 0.13), so the rising share reflects the tighter null distributions of large buyers, that is, statistical power.

Table S9: Incumbency by year.

Year	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
2026	46	0.587	0.079 [0.022, 0.152]	0.151	0.301
2025	353	0.450	0.067 [0.045, 0.093]	0.112	0.222
2024	350	0.557	0.078 [0.054, 0.106]	0.144	0.263
2023	413	0.496	0.098 [0.075, 0.121]	0.131	0.265
2022	463	0.501	0.080 [0.058, 0.102]	0.132	0.260
2021	462	0.450	0.068 [0.050, 0.089]	0.106	0.222
2020	410	0.483	0.076 [0.054, 0.100]	0.131	0.224
2019	307	0.518	0.072 [0.046, 0.101]	0.122	0.262
2018	345	0.441	0.050 [0.029, 0.072]	0.095	0.147
2017	363	0.355	0.046 [0.028, 0.066]	0.082	0.148
2016	320	0.409	0.043 [0.025, 0.066]	0.087	0.174
2015	310	0.410	0.041 [0.023, 0.065]	0.098	0.167
2014	295	0.400	0.035 [0.017, 0.054]	0.091	0.165
2013	234	0.410	0.039 [0.017, 0.064]	0.084	0.148
2012	165	0.352	0.024 [0.006, 0.048]	0.060	0.137
2011	76	0.368	0.022 [0.000, 0.053]	0.118	0.136
2010	1	0.000	0.029 [0.000, 1.000]	0.033	0.000

Table S10: Sensitivity of the incumbency result (1,000 permutations each; all $p < 0.001$).

Variant	Observed	Null mean [95% interval]	Ratio
Main (N1: market $\times$ year)	0.452	0.062 [0.056, 0.067]	7.3
N1 with 3-year windows	0.452	0.060 [0.055, 0.066]	7.5
N2: market $\times$ year $\times$ sector	0.452	0.110 [0.103, 0.116]	4.1
Buyer = recorded label	0.462	0.106 [0.100, 0.112]	4.4
Firm = recorded name	0.394	0.047 [0.042, 0.052]	8.4
Excluding call-centre contracts	0.452	0.062 [0.056, 0.067]	7.3
Including ambiguous scope	0.435	0.056 [0.051, 0.061]	7.8
Cells = buyer sector $\times$ year	0.452	0.047 [0.042, 0.052]	9.7
Incumbency at buyer level (any market)	0.389	0.053 [0.049, 0.058]	7.3
Excluding Article 21(f) contracts	0.495	0.079 [0.071, 0.087]	6.3
N3: market $\times$ year $\times$ province	0.452	0.207 [0.201, 0.213]	2.2

S4 Supplier choice and state dependence

Choice sets. For each repeat-eligible contract, the alternatives are the firms that won at least one contract in the same product market in the 24 months strictly before the tender date; covariates use only information before the tender. In 38.7% of the 4,913 repeat-eligible contracts the winner is not in this set (an entrant to the market or returning after more than 24 months): 65% of non-incumbent wins, 20% of renewals, 49% of new needs, and 52% of 21(f) contracts. The main model uses the remaining 3,013 contracts (909 buyers). At least one of the buyer’s incumbents is in the choice set in 87.5% of these contracts, and the most recent supplier in 85.8%. An earlier version of the choice sets included firms whose only wins came after the tender. That produced an artificial “no prior wins” effect and is not used.

Estimates. Table S12 reports the incumbent odds ratio across specifications. Adding entrant winners back to their choice sets gives a lower bound of 21.5 (18.5–25.0), because an entrant indicator cannot be estimated (it is one only for chosen alternatives). With a pool that includes firms active in any market, the incumbent odds ratio is 31.9 and firms new to the market have an odds ratio of 0.23.

Future-supplier placebo. We add an indicator for firms that supply the buyer in the market after the tender date but had not before. In the main specification its odds ratio is 54.4 (42.2–70.1), next to 63.7 for past suppliers; restricting to tenders up to 2023, to limit right-censoring of future ties,

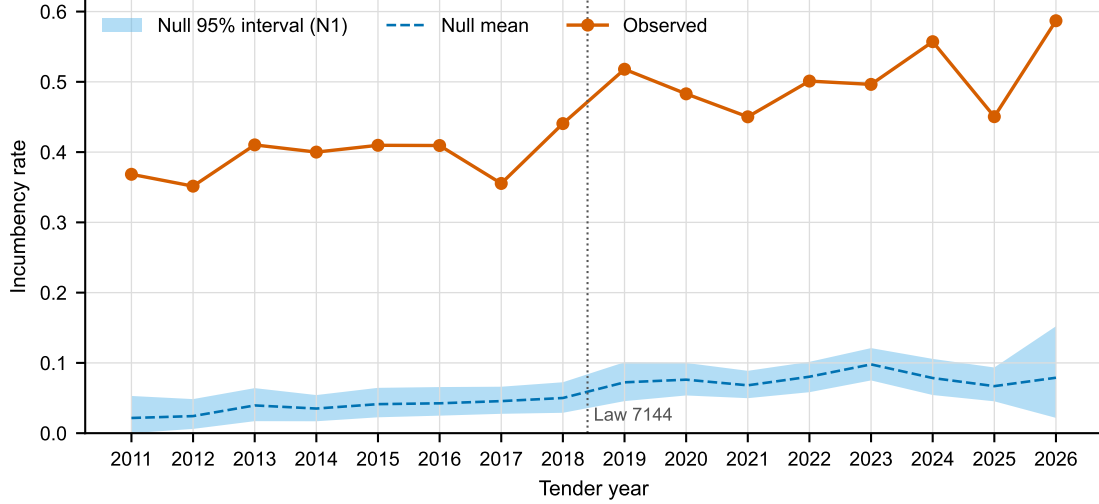


Figure S1: Incumbency rate by year, observed and under null N1 (mean and 95% interval).

Table S11: Incumbency of renewals and new needs: observed [N1 null; N3 null].

	Renewals	New needs
All	0.720 [0.092; 0.288] ($n=1,743$)	0.305 [0.045; 0.163] ($n=3,170$)
Article 21(b)	0.900 [0.198; 0.427] (291)	0.566 [0.093; 0.331] (327)
Open	0.669 [0.077; 0.267] (1,128)	0.307 [0.045; 0.165] (1,913)
Other (mainly 21(f))	0.735 [0.049; 0.237] (324)	0.210 [0.026; 0.099] (930)
Health information systems	0.869 [0.209; 0.452] (519)	0.607 [0.110; 0.375] (733)
ERP / management software	0.821 [0.034; 0.227] (201)	0.233 [0.024; 0.069] (544)
Software licences	0.593 [0.048; 0.212] (241)	0.161 [0.034; 0.091] (361)
Custom software	0.544 [0.010; 0.152] (79)	0.134 [0.006; 0.038] (409)
Maintenance / support	0.776 [0.036; 0.138] (192)	0.284 [0.021; 0.107] (183)
Network / data centre	0.670 [0.035; 0.184] (112)	0.236 [0.016; 0.090] (216)
Computers / peripherals	0.436 [0.014; 0.194] (110)	0.257 [0.010; 0.146] (171)
GIS / city information	0.828 [0.135; 0.274] (58)	0.157 [0.071; 0.104] (172)
Threshold 0.40	0.709 [-; 0.286]	0.296 [-; 0.159]
Threshold 0.67	0.744 [-; 0.302]	0.329 [-; 0.168]
Window 0–18 months	0.695 [-; 0.293]	0.268 [-; 0.143]
Window 0–24 months	0.684 [-; 0.290]	0.250 [-; 0.135]

gives 52.4. The design inflates the future coefficient even without heterogeneity, because winning the current tender makes the winner an incumbent for later tenders. We therefore simulated 100 data sets with state dependence only. Winners of the 3,013 contracts are re-drawn in chronological order from the fitted main model, with incumbency recomputed from the simulated history. All other covariates keep their observed values, and entrant and first contracts keep their observed winners. The placebo model is then re-estimated on each simulated data set. The simulated future-supplier odds ratio is 19.8 (16.1–24.5), far below the observed 54.4, so persistent buyer–firm matching accounts for a substantial part of the incumbency effect. The comparison of past and future suppliers differs by type of purchase. For renewals the odds ratios are 258 and 128, for new needs 30 and 40.

Firm-level versus dyadic heterogeneity. Adding the firm’s future wins in the same market from other buyers (odds ratio 1.40 per log unit) reduces the future-supplier odds ratio from 54.4 to 42.0 (32.3–54.5), and adding past wins from other buyers as well gives 41.5; for renewals it falls from 128 to 96 and for new needs from 40 to 31. Counting only future ties formed more than 30, 90, 180, or 365 days after the tender gives 56.1, 55.3, 52.7, and 48.7 (35.5 at 365 days with the firm controls).

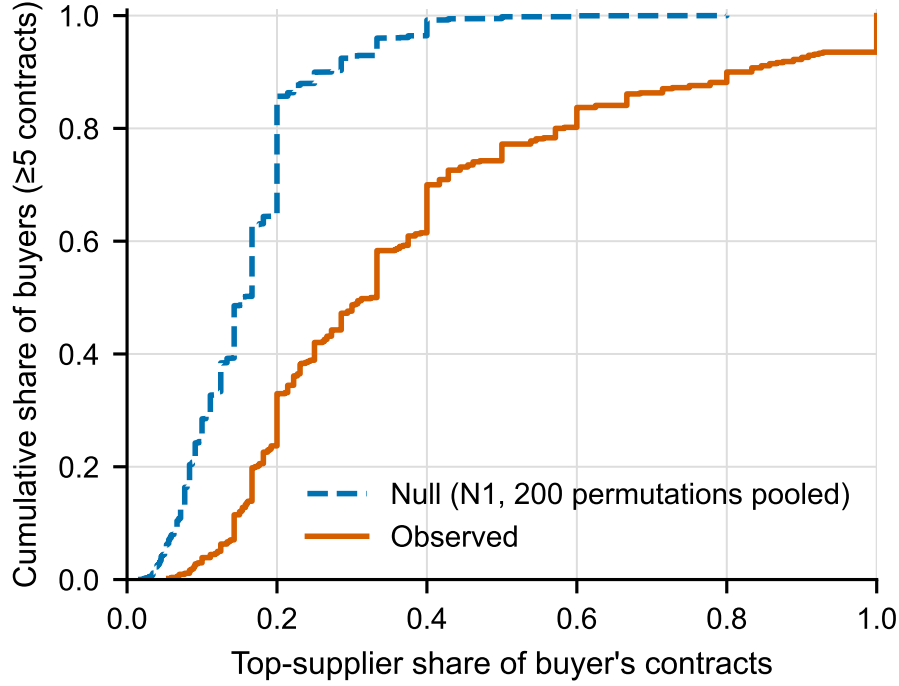


Figure S2: Distribution of the main supplier's share of each buyer's contracts (540 buyers with at least five contracts), observed and under null N1.

S4.1 Simulated worlds

State dependence only. Winners of the 3,013 contracts are re-drawn in chronological order from the fitted model, with incumbency recomputed from the simulated history; entrant and first contracts keep their observed winners. With all other covariates held at observed values the future-supplier odds ratio is 19.8 (16.1–24.5); with every covariate recomputed from the simulated history it is 24.2 (20.0–28.9); with coefficients drawn from their estimated distribution 19.8 (15.3–25.8). Table S13 shows the dependence on the incumbency coefficient. The coefficient that reproduces the observed share of incumbent winners (0.689) is 3.93 (odds ratio 51) and gives a future-supplier odds ratio of 21.9 (17.9–26.5); no value reproduces both the share and the observed future-supplier odds ratio.

Heterogeneity only. The incumbency coefficient is set to zero, and a persistent buyer–firm random intercept with standard deviation σ is calibrated to the observed share of incumbent winners ($\sigma = 2.92$ overall, 4.44 for renewals, 2.47 for new needs). This world also reproduces incumbency among all repeat-eligible contracts (0.457 against 0.452). Table S14 compares the three worlds; Figure 4 of the main text plots them.

The observed past/future ratio is below the state-dependence world in every sample (all $z = -6.6$, renewals -4.9 , new needs -7.9 ; no simulated replication reaches the observed value). Against the heterogeneity world it is lower overall ($z = -2.9$) and for new needs (-2.8) but higher for renewals ($+2.5$). Heterogeneity alone produces a recency ratio above one, because a displaced supplier has revealed a worse match, so the last-versus-earlier contrast (1.69, 95% CI 1.38–2.07, with a control for the number of past wins with the buyer) does not discriminate between the mechanisms.

Intermediate worlds. Fixing the incumbency coefficient at 0, 1, 2, or 3 and recalibrating σ each time (50 replications), renewals fit between coefficients 2 and 3 (odds ratios 7–20): future-supplier odds ratio 183 and 125, past/future ratio 1.83 and 2.55, recency 2.35 and 2.00, against observed 128, 2.02, and 1.75. New needs fit best with no state dependence; their future-supplier odds ratio exceeds even the heterogeneity world, which suggests forward-looking ties such as multi-stage projects, or misspecification. The pooled sample fits no single combination because it mixes the two groups.

Table S12: Conditional logit: incumbent odds ratio [buyer-clustered 95% CI].

Specification	Contracts	Incumbent OR
24-month pool, all active firms (main)	3,013	42.3 [36.4, 49.2]
24-month pool, 30 random alternatives	3,013	41.8 [35.8, 48.8]
24-month pool, 10 random alternatives	3,013	44.5 [36.3, 54.5]
12-month pool, all active firms	2,456	39.8 [33.8, 46.9]
Renewals	1,388	140.7 [107.0, 185.1]
New needs	1,625	21.0 [17.6, 24.9]
Article 21(b)	529	95.8 [69.4, 132.3]
Open	1,877	33.7 [28.1, 40.5]
Other procedures (mainly 21(f))	607	43.7 [32.3, 59.1]
Entrant winners added (30 alternatives; lower bound)	4,913	21.5 [18.5, 25.0]
Renewals / new needs		48.1 / 13.0

Table S13: State-dependence world: moments by incumbency coefficient (other coefficients at fitted values; 100 replications).

β_{inc} (odds ratio)	Share of incumbent winners	Future-supplier OR [95%]	Past/future ratio	Last/earlier ratio
2.745 (15.6)	0.510	10.0 [7.8, 12.3]	2.02	1.01
3.245 (25.7)	0.590	14.3 [11.4, 17.9]	2.36	0.99
3.745 (42.3), fitted	0.665	19.8 [16.1, 24.5]	2.88	1.01
3.93 (51), calibrated	0.691	21.9 [17.9, 26.5]	3.19	1.00
4.245 (69.8)	0.730	26.6 [20.2, 33.7]	3.62	1.00
4.745 (115.0)	0.782	35.1 [27.8, 45.1]	4.56	1.01
Observed	0.689	54.4 [42.2, 70.1]	1.17	1.75

S5 Procedure and timing

S5.1 Contract-level model

The 21(b) $\times$ post-2018 odds ratio is 2.42 (1.43–4.09) with the buyer-history controls and two-way clustering, and 2.42 (1.57–3.73) with buyer clustering. Without the prior-contract and prior-supplier controls it is 2.90 (1.64–5.15); without any buyer-history controls it is 2.93 (1.65–5.19). The pre-2018 marginal effect of 21(b) is between -3.5 and -3.8 percentage points and insignificant in all specifications. Further checks give interaction odds ratios between 1.76 and 2.43: an IT-only sample, broad scope, recorded buyer labels, recorded firm names, a linear trend instead of year fixed effects, and excluding 2020. Alternative break dates give 2.55 (25 August 2017, KHK 694; log-likelihood -2801.1 against -2802.5 for 25 May 2018), 2.10 (placebo, May 2016, $p = 0.009$), and 1.57 (placebo, May 2020, $p = 0.08$). The interaction is 2.64 among health buyers and 0.86 ($p = 0.78$) among other buyers. In a model with a triple interaction (two-way clustered), the post-2018 21(b) interaction is 2.44 ($p = 0.003$) for health buyers and 0.89 for others. The difference is not significant (odds ratio 2.74, 95% CI 0.79–9.51, $p = 0.11$; in percentage points $+13.9$, -7.6 to 35.5), because only 84 non-health 21(b) contracts are repeat-eligible. Under Article 21(f), 34.6% of repeat-eligible contracts go to an incumbent, against 13.1% under N3 and 3.3% under N1.

S5.2 Event study

Figure S3 reports the 21(b)–open gap by year relative to 2013–17. In that period the gap does not differ across years (joint tests $p = 0.19$ for all buyers and 0.20 for health buyers). In 2018 it is 37 percentage points higher for all buyers (24–50) and 43 points higher for health buyers (23–62). From 2019 on it averages 14 points above the reference for all buyers ($p = 0.003$) and 12 points for health buyers ($p = 0.04$). With buyer fixed effects, health buyers show a gap averaging -10 points in 2013–17, positive in every year from 2018, and 24 points (10–38) above the reference from 2019 on.

Table S14: Observed moments (95% CI) and simulated worlds (mean [2.5–97.5%], 100 replications).

Sample	Moment	Observed	State dependence only	Heterogeneity only
All	Future-supplier OR	54.4 (42.2–70.1)	19.8 [16.1, 24.5]	40.5 [30.6, 51.5]
	Past/future ratio	1.17 (0.95–1.44)	2.88 [2.48, 3.37]	1.82 [1.45, 2.22]
	Last/earlier ratio	1.75 (1.43–2.14)	1.01 [0.85, 1.17]	2.29 [1.92, 2.60]
Renewals	Future-supplier OR	128 (83–197)	23.6 [14.1, 38.4]	341 [207, 504]
	Past/future ratio	2.02 (1.43–2.85)	8.18 [5.29, 12.24]	1.14 [0.84, 1.51]
	Last/earlier ratio	1.75 (1.34–2.29)	1.01 [0.83, 1.24]	2.99 [2.33, 3.60]
New needs	Future-supplier OR	39.9 (28.8–55.2)	7.2 [5.4, 9.5]	21.7 [15.7, 28.3]
	Past/future ratio	0.76 (0.57–1.01)	3.44 [2.62, 4.22]	1.39 [1.04, 1.82]
	Last/earlier ratio	1.60 (1.21–2.11)	1.00 [0.79, 1.22]	1.63 [1.30, 2.02]

Among parametric shapes, a step in August 2017 combined with a declining trend fits best. A step in May 2018 fits worse by 3.4–7.8 AIC points across the three samples, and a pure trend fits about as badly as no change. Only 8 and 10 repeat-eligible 21(b) contracts fall in the windows 25 August–31 December 2017 and 1 January–24 May 2018. The years 2010–12, with little buyer history and 24 21(b) contracts, show a high gap as well. Against that base, no later year is significantly higher.

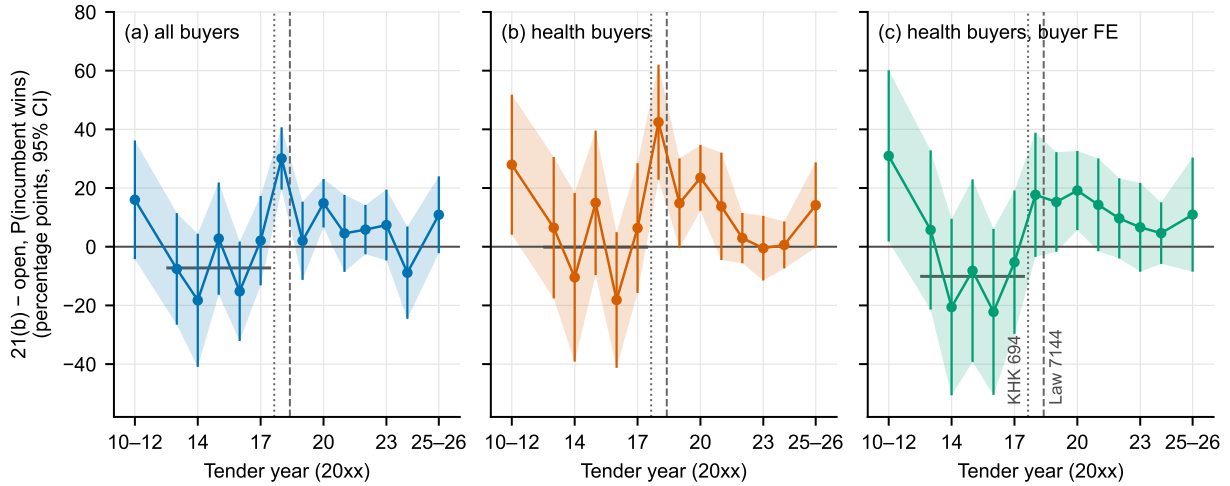


Figure S3: Difference in the probability that an incumbent wins between Article 21(b) and open tenders, by year. Linear probability models with product-market, year, and buyer-sector fixed effects (buyer fixed effects in panel c), log value, and buyer-history controls; two-way clustered 95% intervals. Horizontal bars: 2013–17 reference mean. Dotted and dashed lines: KHK 694 (25 August 2017) and Law 7144 (25 May 2018).

S5.3 Dyad continuation

For each new buyer–firm pair in a product market, we ask whether the pair signs another contract in that market. Of 7,761 pairs, 4,334 have no later contract by the buyer in that market and cannot continue. The analysis uses the remaining 3,427, of which 34.6% continue. Covariates are measured at the first contract without look-ahead. The number of later opportunities (the buyer’s later contracts in the market) enters as a covariate, because the offset restriction is rejected (coefficient 0.90 in Poisson pseudo-maximum likelihood, $p = 0.02$; 0.88 in NB2, $p = 0.006$). In NB2 the dispersion parameter is estimated ($\alpha = 0.46$, SE 0.12; likelihood-ratio statistic for $\alpha = 0$: 152.5). Firm size and prior distinct buyers are highly collinear ($r = 0.99$; VIFs of 63 and 61), and generalism correlates 0.86 with size, so these variables are not separately identified in the full model. Entered alone, generalism and size are both positively associated with continuation. There is no evidence

that specialist firms are more likely to keep a buyer. A first contract under 21(b) is associated with continuation (odds ratio 1.85).

Table S15: Dyad continuation, logit (full specification; product-market and year fixed effects; two-way clustered 95% CIs).

	OR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.85 [1.37, 2.50]	<0.001
First contract other negotiated (vs open)	1.26 [1.03, 1.54]	0.025
Log real value of first contract	1.02 [0.95, 1.10]	0.552
Log firm prior contracts (other buyers)	1.58 [0.97, 2.58]	0.067
Log firm prior markets (generalism)	1.20 [0.86, 1.66]	0.284
Log firm prior distinct buyers	0.69 [0.41, 1.15]	0.156
Log buyer prior contracts	0.76 [0.69, 0.85]	<0.001
Log later opportunities of buyer in market	1.89 [1.68, 2.13]	<0.001

Table S16: Number of later contracts, NB2 with estimated dispersion (full specification).

	IRR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.61 [1.39, 1.87]	<0.001
First contract other negotiated (vs open)	1.19 [1.02, 1.39]	0.029
Log real value of first contract	1.01 [0.96, 1.06]	0.691
Log firm prior contracts (other buyers)	1.36 [1.07, 1.74]	0.012
Log firm prior markets (generalism)	1.20 [0.98, 1.45]	0.074
Log firm prior distinct buyers	0.76 [0.59, 0.99]	0.045
Log buyer prior contracts	0.77 [0.72, 0.82]	<0.001
Log later opportunities of buyer in market	2.42 [2.23, 2.63]	<0.001

S6 Single bidding

Random sample. Bid counts were retrieved from published result announcements for a random sample of tenders stratified by year (40 per year drawn from all retrieved tenders). Of these, 393 are in the main sample and report bid counts; none is available for 2010–2011. Rates are weighted by the inverse sampling fraction of each year, with linearized 95% intervals. Incumbency status is undefined when the tender is the buyer’s first in the market. Adjusting for procedure and year (logit, buyer-clustered, $n = 208$ tenders with defined incumbency), the odds ratio of a single bid for incumbent winners is 4.44 (2.39–8.25). Adding renewal status as a control gives odds ratios of 3.62 for a single bid and 2.95 for a single valid bid (2.63 without the renewal control). The single-bid rate has a weak upward trend (odds ratio 1.05 per year; $p = 0.04$ unweighted, $p = 0.11$ weighted). In 308 tenders with a recorded number of firms that obtained the documents, design-weighted averages are 4.0 such firms and 1.9 valid bids. In 45% of these tenders two or more firms obtained the documents but only one valid bid was received.

Software sample. An interrupted enrichment run of the scraper retrieved result details for 3,400 tenders found with the first search keyword, *yazılım* (software). For tenders from 2021 on, these include the number of valid bids. Joined to the main sample, 1,194 tenders have this count, about 94% of software-titled tenders from 2021 on but only 35% of all tenders in that period. The sample is therefore not random. Single valid bids occur in 66.7% of incumbent-won tenders and 49.3% of others. In a logit with procedure, year, and market fixed effects (681 tenders with defined incumbency, buyer-clustered), the incumbent odds ratio is 2.04 (1.40–2.96), and 2.68 when renewal status is added (renewal odds ratio 0.52). Controlling for renewal raises the incumbent effect because renewals are much more common among incumbent winners (55% against 13%), and renewals won by newcomers are the most contested tenders (29% single valid bids, against 52% for new needs won by newcomers).

Table S17: Single-bid rates (design-weighted random sample).

Group	Tenders	Single bid, % [95% CI]	Single valid bid, %
All tenders	393	44.2 [39, 49]	55.8
Open	225	40.8 [34, 48]	55.9
Article 21(b)	50	75.0 [60, 86]	81.6
Article 21(f)	114	40.0 [31, 50]	46.9
Other negotiated	4	0.0	18.5
Winner is incumbent	95	71.2 [61, 80]	76.7
Winner not previous supplier in market	113	32.4 [24, 42]	48.4
Buyer's first contract in market	185	38.3 [31, 46]	50.2
Before 25 May 2018	140	38.6 [31, 47]	49.1
After 25 May 2018	253	48.9 [42, 55]	61.5

S7 Estimated costs and discounts

Sample. From the 4,913 repeat-eligible contracts, 75 won by incumbents and 75 by other firms were drawn at random (seed 11). An AI agent opened each tender's detail page on EKAP in the authors' ordinary browser session, at a pace of roughly one tender per 40 seconds with pauses when the site slowed. It recorded the estimated cost, contract value, highest and lowest bids, product codes, and bid counts, without names, and its input list did not show incumbency. It did not interact with any verification widget. It collected 144 tenders before the site displayed a human-verification check, at which point collection stopped. One contract was in US dollars; 143 have both values in lira. EKAP contract values match our records within 1% for 99.3% of matched tenders.

Discounts. The discount is $1 - \text{contract value} / \text{estimated cost}$. It averages 14.2% (median 10.4%, range -9.9% to 84.6%).

Table S18: Discount on the estimated cost by winner type.

Group	n	Mean	Median	Zero discount
Incumbent winner	73	0.121	0.089	0.082
Other winner	70	0.165	0.112	0.043
Renewal, incumbent	46	0.120	0.087	0.043
Renewal, other	9	0.156	0.019	0.000
New need, incumbent	27	0.123	0.101	0.148
New need, other	61	0.166	0.132	0.049

In OLS with robust standard errors, the incumbent coefficient is -0.044 (SE 0.027, $p = 0.11$) without controls, -0.060 (0.038, $p = 0.11$) with renewal, procedure, log value, and year, and -0.050 (0.041, $p = 0.22$) with product-market fixed effects as well. Bid counts on the detail pages are available mainly for recent tenders (68 of 144); in that small subsample incumbent-won tenders have 1.8 valid bids on average against 1.3 for others, the opposite of the pattern in the larger bid samples, and we do not interpret it.

Product codes. Among the 143 tenders, the first listed OKAS (CPV) code is in division 72 (IT services) for most software markets and 48 (software packages) or 50 (repair and maintenance) for licence and maintenance contracts. Division-level codes do not discriminate between our product markets, which are finer.

S8 Network structure

Graph. The main-sample graph has 2,814 firms, 2,890 buyers and 7,219 weighted edges, with density 8.9×10^{-4} . The giant component contains 4,446 nodes (78%) and 92% of contracts. Degree distributions (number of distinct partners) are heavy-tailed. Fitted by maximum likelihood with the protocol of Clauset, Shalizi and Newman (2009), firms have $\hat{\alpha} = 2.21$, and a lognormal is not

distinguishable from a power law ($R = -1.67$, $p = 0.09$). For buyers ($\hat{\alpha} = 2.06$) the power law is rejected by the bootstrap goodness-of-fit test, and a lognormal fits better ($R = -6.19$, $p < 0.001$). Degree assortativity is -0.17 against -0.04 in the null. Bipartite clustering exceeds the null: Robins–Alexander 0.026 against 0.012, Latapy 0.31 against 0.25. Figure S4 shows the core of the network.

Communities. Over 50 Louvain runs with random seeds and node orders, weighted modularity of the giant component ranges from 0.760 to 0.764 (34–43 communities). Barber’s bipartite modularity of the same partitions is practically identical. In 200 bipartite degree-preserving randomizations (curveball), the unweighted modularity of the giant component is 0.694 ± 0.003 against an observed 0.731, and the weighted modularity is 0.756 ± 0.003 against an observed 0.761. Communities align with buyer sectors (NMI 0.25; permuted labels 0.03; purity 0.64 against a largest-label baseline of 0.44), and ten communities consist of at least 91% health buyers. They align only weakly with firms’ main product markets (NMI 0.13; permuted 0.04). In annual full graphs, modularity exceeds its bipartite null only modestly (annual z between 0.4 and 3.9). Within annual giant components it does not exceed the null in 16 of 17 years, so we do not interpret yearly modularity as evidence of persistent segmentation.

Projections and brokerage. We computed exact Poisson-binomial p -values for co-occurrences under the bipartite configuration model (BiCM). With a false-discovery-rate correction over all pairs at 0.05 or 0.01, no link is validated in either projection, because the data are sparse: 94–97% of co-occurring pairs share exactly one neighbour. A liberal variant that corrects only over co-occurring pairs retains 15,473 firm links and 42,005 buyer links. It essentially reproduces the raw projection without hub-mediated links, so we do not treat it as a validated backbone. Disparity-filter backbones of Newman-weighted projections at $\alpha = 0.05$ contain 1,077 firm links (modularity 0.73) and 2,685 buyer links (0.73). They overlap little with the liberal BiCM set (Jaccard 0.04).

Across these projections and the raw projection, firm betweenness correlates with the number of product markets a firm serves, but also with its number of buyers. Controlling for buyers and contracts, the partial Spearman correlation is 0.12–0.15, and adding the number of markets to a size-only regression raises R^2 by 0.003–0.015. Brokerage thus adds little beyond firm size. In the raw buyer projection, the most connected buyer, a provincial health directorate once pooled labels are split, shares at least one supplier with 19% of the projection’s giant component. With recorded pooled labels the figure is 45%, which shows how strongly such statistics depend on buyer coding.

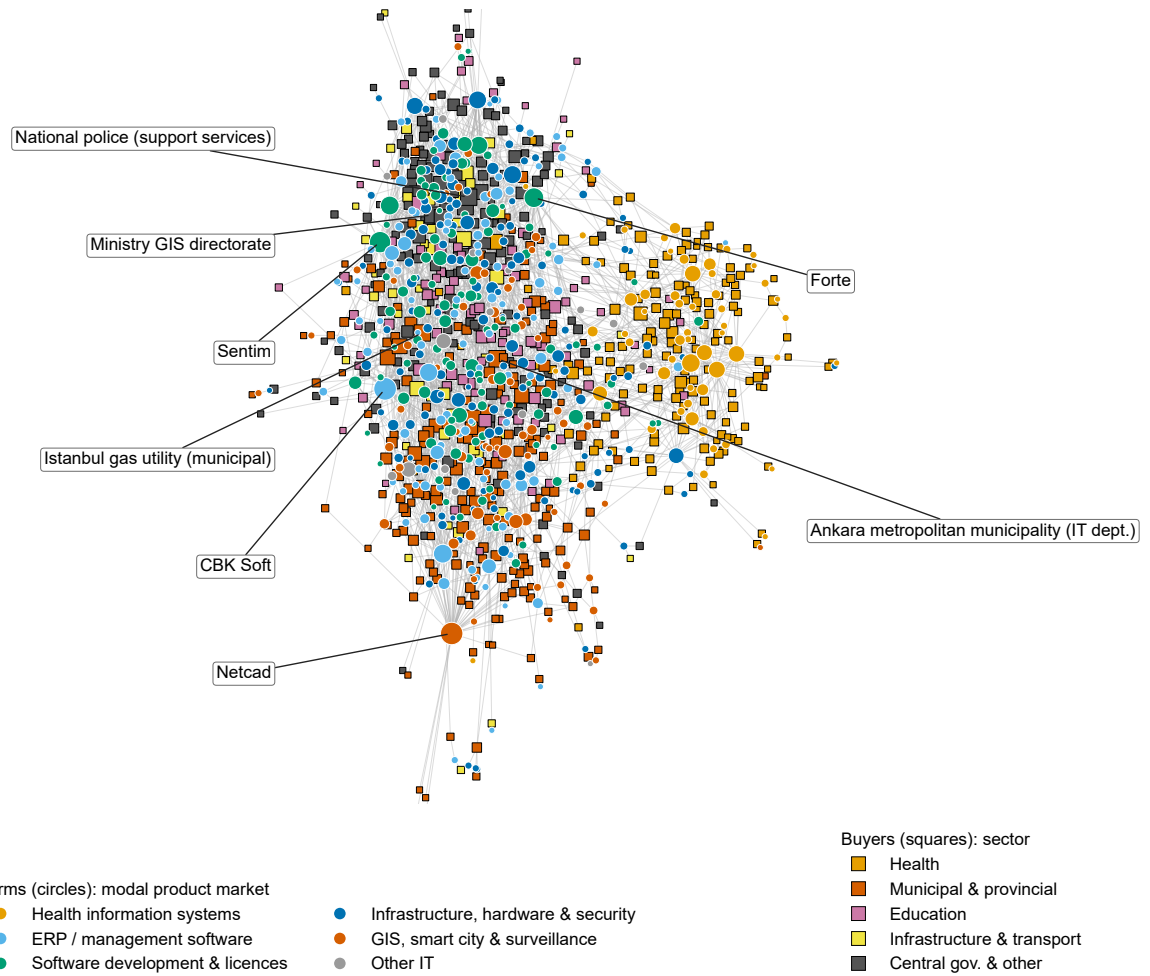


Figure S4: Core of the buyer-supplier network (nodes of degree ≥ 3 , largest connected component; force-directed layout with fixed seed). Circles are firms coloured by their main product market; squares are buyers coloured by sector.